Department of Metallurgical and Materials Engineering
Indian Institute of Technology Kharagpur
Final examination, Spring Semester, 2024

Time: 3 hrs. MT20202– Materials Processing Marks: 8x10=80

(Answer all the questions)

1. (a) Although both are powder-bed based additive manufacturing, explain the difference between ceramic 3d printing and SLM of metal. 6+4=10

(b) Given similar press conditions, green density of ceramics are generally lower than metals (similar particle size and distribution). Explain

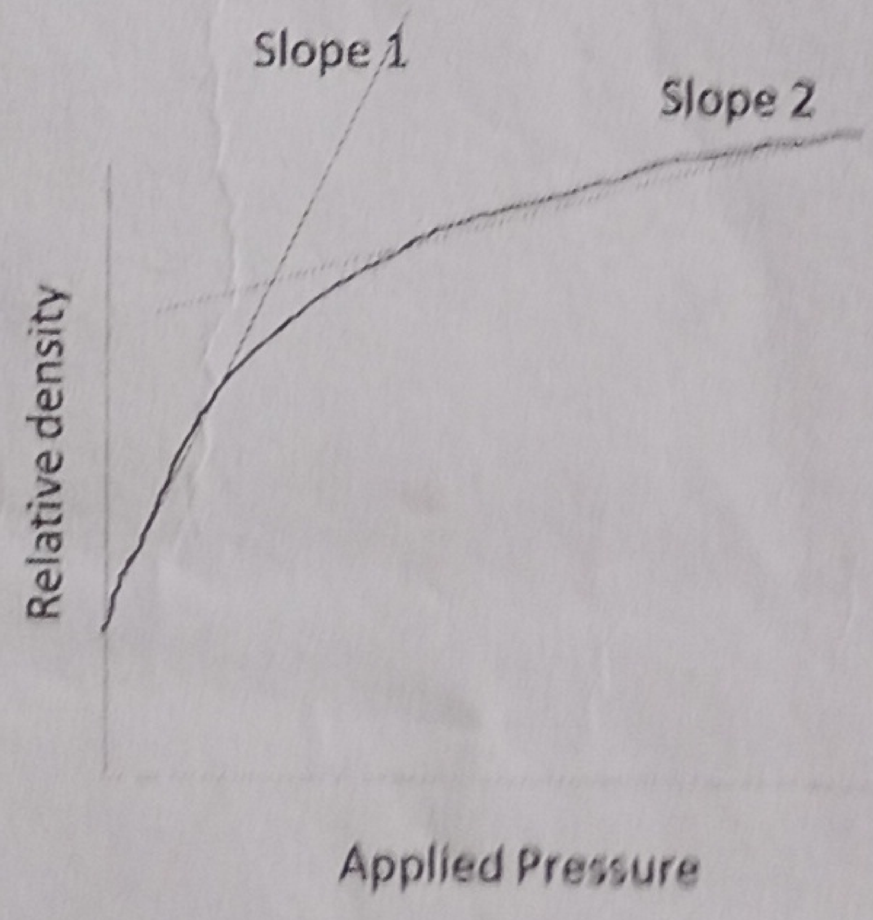
2. With schematics, briefly explain how you will develop nanometric sized structure in MEMS. What is the basic difference between CVD and PVD process? 7+3 =10

3. Give 2 reasons why you will select powder metallurgy over melting casting and vice versa. Explain how some diffusions lead to densification while the others don't. 6+4= 10

4. Graphically represent uniumodal, bimodal, and polymodal particle size distribution. In a single graph show narrow and wide distributions of 149 μm sized particles. What do you understand by D_{10}, D_{50}, and D_{90} in powder size distribution? 3+3+4= 10

5. In a single action uniaxial compaction, green density decreases along the height of the compact. Explain with schematics? Also explain why this is a concern and how to avoid it?10

6. (a) The following graph shows green density vs. compaction pressure. Explain the reason for two distinctive slopes? 7+3=10



(b) Mention 3 full density processing techniques.

7. (a)In two cups of tea (100 ml liquid), you have added 1 spoon (fixed volume) of (a) powdered ultrafine sugar and (b) granulated sugar of same brand and quality. Explain whether there will be any change in sweetness in the two cups of tea. Explain yourself.

(b) How a pore gets eliminated in solid state sintering 6+4=10

8. What are the three broad classification of powder production processes? How do we achieve powder using mechanical methods. 3+7=10